1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t/\gamma_N = a_o R$ [μm/pN]; $\bar{a} := a/a_{\text{nom}}$; forces by a_o: $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $0 \le \bar{a} \le 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{\bar{a}}_w$, $\hat{b}$, $\delta\hat{\bar{w}}$; $e_\theta = \theta - \hat{\theta}$)

$$
\begin{aligned}
\bar{a}(\bar{w}) &= 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)} \\
\bar{a}'(\bar{a}, b) &:= \frac{d\bar{a}}{d\bar{w}} = b\big(1 - \bar{a}\big)^2 \\
\bar{a}''(\bar{a}, b) &:= \frac{d\bar{a}'}{d\bar{w}} = -2\,b^2\big(1 - \bar{a}\big)^3 \\
a_w &= a_o\,\bar{a}_w, \qquad a_o = \frac{\Delta t}{\gamma_N R}
\end{aligned}
$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + b[k]\big(1 - \bar{a}_w[k]\big)^2\,\Delta\bar{w}[k] \\
b[k+1] &= b[k]
\end{aligned}
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1 - \lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1 - \lambda_c)\,n_w[k]
\end{aligned}
$$

3 Estimator

$$
\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top \qquad (d = 2)
$$

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}
$$

4 Error dynamics

(True − Estimator; coefficients at the estimate)

$$
\begin{aligned}
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
b[k]\big(1-\bar{a}_w[k]\big)^2 &= \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2 - 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\,e_{\bar{a}_w}[k] + \big(1-\hat{\bar{a}}_w[k]\big)^2 e_b[k] + \mathcal{O}\big(e^2\big)
\end{aligned}
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big[1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\Big]e_{\bar{a}_w}[k] \\
&\quad + (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \big(1-\hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)e_b[k] + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w[k] \\
e_b[k+1] &= e_b[k]
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw} & 0 \\
0 & 0 & (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & \begin{array}{l}1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} \\ \quad -2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\end{array} & \begin{array}{l}\big(1-\hat{\bar{a}}_w\big)^2 \\ \times\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\end{array} \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \Big[\,0,\ 0,\ -\varepsilon_w,\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w,\ 0\,\Big]^\top
$$

5 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]\big)$

$$
\begin{aligned}
y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
\hat{y}_1[k] &= \delta \hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
H_{24}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{\bar{a}}_w[k]} = 1 + 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\,\nabla_d \bar{w}_d[k] \\
H_{25}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{b}[k]} = -\big(1-\hat{\bar{a}}_w[k]\big)^2 \nabla_d \bar{w}_d[k] \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{24}[k] & H_{25}[k] \end{bmatrix} \\
\mathbf{e}_y[k] &= H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}
\end{aligned}
$$

6 Noise

$$
\begin{aligned}
Q[k] &= Q_{33}[k]\,\mathbf{g}[k]\,\mathbf{g}[k]^\top, \qquad \mathbf{g}[k] = \Big[\,0,\ 0,\ -1,\ \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2,\ 0\,\Big]^\top, \qquad Q_{55} = 0 \\
Q_{33}[k] &= \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_B T\,a_o}{R} \\
R[k] &= \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2
\end{aligned}
$$

7 Observability

$(\mathbf{e}[k] = [\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_b \,]^\top$, $\mathbf{n}[k] = [\, n_w[k],\ n_a[k] \,]^\top$
$\Phi(k, k_0) := F_e[k-1]\, F_e[k-2] \cdots F_e[k_0]$, $\Phi(k_0, k_0) = I$
N window length
$\mathbf{Y} := [\, \mathbf{e}_y[k_0];\ \mathbf{e}_y[k_0+1];\ \ldots;\ \mathbf{e}_y[k_0+N] \,]$, $\mathbf{N} := [\, \mathbf{n}[k_0];\ \mathbf{n}[k_0+1];\ \ldots;\ \mathbf{n}[k_0+N] \,]$
$X_{:,j}$ column j of X)

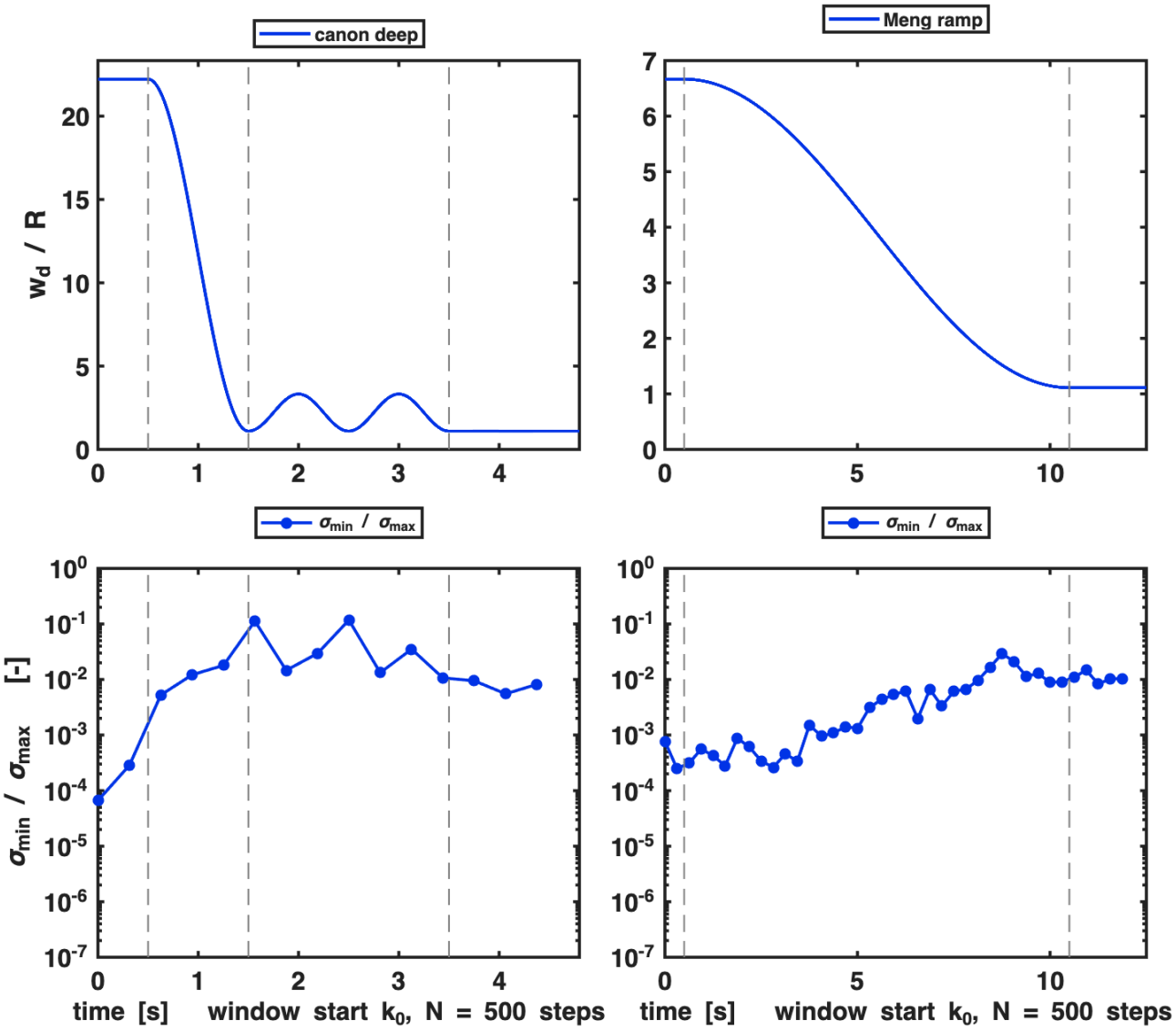
$$F_e[k]_{:,5} = \Big[\, 0,\ 0,\ 0,\ \big(1-\hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big),\ 1 \,\Big]^\top$$
$$H[k]_{:,5} = \Big[\, 0,\ -\big(1-\hat{\bar{a}}_w\big)^2\,\nabla_d\bar{w}_d \,\Big]^\top$$

$$\mathbf{e}[k] = \Phi(k, k_0)\,\mathbf{e}[k_0]$$
$$\mathbf{e}_y[k] = H[k]\,\Phi(k, k_0)\,\mathbf{e}[k_0] + \mathbf{n}[k]$$
$$\mathbf{Y} = O(k_0, N)\,\mathbf{e}[k_0] + \mathbf{N}, \qquad O(k_0, N) := \begin{bmatrix} H[k_0]\,\Phi(k_0, k_0) \\ H[k_0+1]\,\Phi(k_0+1, k_0) \\ \vdots \\ H[k_0+N]\,\Phi(k_0+N, k_0) \end{bmatrix}$$

$$R_w := \operatorname{diag}\big(R[k_0],\ R[k_0+1],\ \ldots,\ R[k_0+N]\big), \qquad O_w(k_0, N) := R_w^{-1/2}\,O(k_0, N)$$
$$\mathbf{Y}_w := R_w^{-1/2}\,\mathbf{Y} = O_w(k_0, N)\,\mathbf{e}[k_0] + \mathbf{N}_w, \qquad \mathbb{E}\big[\mathbf{N}_w\mathbf{N}_w^\top\big] = I$$

$$O_w(k_0, N)\,\mathbf{v}_i = \sigma_i\,\mathbf{u}_i, \qquad i = 1, \ldots, 5, \qquad \sigma_1 \geq \cdots \geq \sigma_5 \geq 0, \qquad \|\mathbf{v}_i\| = \|\mathbf{u}_i\| = 1$$
$$\sigma_1 = \max_{\|\mathbf{v}\|=1} \|O_w(k_0, N)\,\mathbf{v}\|, \qquad \sigma_5 = \min_{\|\mathbf{v}\|=1} \|O_w(k_0, N)\,\mathbf{v}\|$$
$$\operatorname{rank} O_w(k_0, N) = \#\{\, i : \sigma_i > 0 \,\}$$
$F_e[k]_{:,j} = I_{:,j}$, $H[k]_{:,j} = 0\ \ \forall k\ \Rightarrow\ O_w(k_0, N)_{:,j} = 0\ \Rightarrow\ \sigma_5 = 0$

$$\hat{\mathbf{e}}[k_0] = \sum_{i=1}^{5} \frac{\mathbf{u}_i^\top \mathbf{Y}_w}{\sigma_i}\,\mathbf{v}_i$$
$$\mathbf{v}_i^\top\big(\hat{\mathbf{e}}[k_0] - \mathbf{e}[k_0]\big) = \frac{\mathbf{u}_i^\top \mathbf{N}_w}{\sigma_i}, \qquad \mathbb{E}\Big[\Big(\mathbf{v}_i^\top\big(\hat{\mathbf{e}}[k_0] - \mathbf{e}[k_0]\big)\Big)^2\Big] = \frac{1}{\sigma_i^2}$$
$\mathbf{v}_5 \to \big[\, 0,\ 0,\ 0,\ 0,\ 1 \,\big]^\top\ \Rightarrow\ \mathbb{E}\big[(\hat{e}_b - e_b)^2\big] \to \dfrac{1}{\sigma_5^2}$
$$\frac{1}{\sigma_5} \lessapprox \sqrt{P_{55}[0]}, \qquad \sqrt{P_{55}[0]} = \sup_{\bar{w}\in\text{env}} \Big|\, b_{\text{true}}(\bar{w}) - \tfrac{8}{9} \,\Big|$$


canon deep
Meng ramp
w_d / R
σ_{min} / σ_{max}
σ_{min} / σ_{max} [-]
time [s] window start k_0, N = 500 steps
time [s] window start k_0, N = 500 steps